

Practice Assignment:

Datasbase:

Part 1:

```
CREATE TABLE customers (  
    id INT PRIMARY KEY AUTO_INCREMENT,  
    name VARCHAR(100),  
    email VARCHAR(100) UNIQUE,  
    signup_date DATE  
);
```

```
INSERT INTO customers (name,email,signup_date) VALUES  
(  
'Aarav Patel','aarav@gmail.com','2024-01-10'),  
(  
'Priya Sharma','priya@gmail.com','2024-01-15'),  
(  
'Rohan Mehta','rohan@gmail.com','2024-02-05'),  
(  
'Sneha Verma','sneha@gmail.com','2024-02-20'),  
(  
'Aditya Singh','aditya@gmail.com','2024-03-01'),  
(  
'Neha Gupta','neha@gmail.com','2024-03-10'),  
(  
'Rahul Joshi','rahul@gmail.com','2024-03-18'),  
(  
'Anjali Shah','anjali@gmail.com','2024-04-05'),  
(  
'Karan Desai','karan@gmail.com','2024-04-10'),  
(  
'Ishita Kapoor','ishita@gmail.com','2024-04-15'),  
(  
'Manav Yadav','manav@gmail.com','2024-05-01'),  
(  
'Pooja Nair','pooja@gmail.com','2024-05-12'),  
(  
'Harsh Patel','harsh@gmail.com','2024-05-20'),  
(  
'Nikhil Jain','nikhil@gmail.com','2024-06-01'),  
(  
'Simran Kaur','simran@gmail.com','2024-06-15');
```

```
CREATE TABLE restaurants (  
    id INT PRIMARY KEY AUTO_INCREMENT,
```

```
name VARCHAR(100),  
cuisine VARCHAR(50)  
);
```

```
INSERT INTO restaurants (name,cuisine) VALUES  
('Pizza Hub','Italian'),  
('Burger Point','Fast Food'),  
('Spice Villa','Indian'),  
('Dragon Bowl','Chinese'),  
('Taco Fiesta','Mexican'),  
('Sushi House','Japanese'),  
('BBQ Nation','Barbecue'),  
('Cafe Delight','Cafe');
```

```
CREATE TABLE accounts (  
id INT PRIMARY KEY AUTO_INCREMENT,  
name VARCHAR(50),  
balance DECIMAL(10,2)  
);
```

```
INSERT INTO accounts (name,balance) VALUES  
('Deepak',5000),  
('Sonu',3000);
```

```
CREATE TABLE orders (  
id INT PRIMARY KEY AUTO_INCREMENT,  
customer_id INT,  
restaurant_id INT,  
total DECIMAL(8,2),  
status VARCHAR(20),  
created_at DATETIME,  
FOREIGN KEY (customer_id) REFERENCES customers(id),  
FOREIGN KEY (restaurant_id) REFERENCES restaurants(id)
```

);

INSERT INTO orders (customer_id,restaurant_id,total,status,created_at) VALUES

(1,1,450,'completed','2025-01-01 12:00:00'),
(2,2,320,'completed','2025-01-02 13:00:00'),
(3,3,600,'pending','2025-01-03 14:00:00'),
(4,4,280,'completed','2025-01-04 15:00:00'),
(5,5,350,'cancelled','2025-01-05 16:00:00'),
(6,6,750,'completed','2025-01-06 17:00:00'),
(7,7,820,'completed','2025-01-07 18:00:00'),
(8,8,290,'pending','2025-01-08 19:00:00'),
(9,1,180,'completed','2025-01-09 12:30:00'),
(10,2,520,'completed','2025-01-10 13:30:00'),
(1,2,340,'completed','2025-01-11 11:00:00'),
(2,3,610,'pending','2025-01-12 12:00:00'),
(3,4,270,'completed','2025-01-13 13:00:00'),
(4,5,460,'completed','2025-01-14 14:00:00'),
(5,6,780,'cancelled','2025-01-15 15:00:00'),
(6,7,890,'completed','2025-01-16 16:00:00'),
(7,8,250,'pending','2025-01-17 17:00:00'),
(8,1,330,'completed','2025-01-18 18:00:00'),
(9,2,560,'completed','2025-01-19 19:00:00'),
(10,3,430,'completed','2025-01-20 20:00:00'),
(11,4,390,'completed','2025-01-21 12:00:00'),
(12,5,620,'pending','2025-01-22 13:00:00'),
(1,6,275,'completed','2025-01-23 14:00:00'),
(2,7,480,'completed','2025-01-24 15:00:00'),
(3,8,710,'completed','2025-01-25 16:00:00'),
(4,1,910,'completed','2025-01-26 17:00:00'),
(5,2,260,'completed','2025-01-27 18:00:00'),
(6,3,340,'pending','2025-01-28 19:00:00'),
(7,4,550,'completed','2025-01-29 20:00:00'),
(8,5,430,'completed','2025-01-30 11:30:00'),
(9,6,315,'completed','2025-02-01 12:30:00'),

```
(10,7,610,'cancelled','2025-02-02 13:30:00'),  
(11,8,290,'completed','2025-02-03 14:30:00'),  
(12,1,410,'completed','2025-02-04 15:30:00');
```

Part 2:

Task 1: Transactions

Schema SQL

```
83 (11,4,390,'completed','2025-01-21 12:00:00'),
84 (12,5,620,'pending','2025-01-22 13:00:00'),
85 (1,5,275,'completed','2025-01-23 14:00:00'),
86 (2,7,480,'completed','2025-01-24 15:00:00'),
87 (3,8,710,'completed','2025-01-25 16:00:00'),
88 (4,1,910,'completed','2025-01-26 17:00:00'),
89 (5,2,260,'completed','2025-01-27 18:00:00'),
90 (6,3,340,'pending','2025-01-28 19:00:00'),
91 (7,4,550,'completed','2025-01-29 20:00:00'),
92 (8,5,430,'completed','2025-01-30 11:30:00'),
93 (9,6,315,'completed','2025-02-01 12:30:00'),
94 (10,7,610,'cancelled','2025-02-02 13:30:00'),
95 (11,8,290,'completed','2025-02-03 14:30:00'),
96 (12,1,410,'completed','2025-02-04 15:30:00');
```

Query SQL

```
1 START TRANSACTION;
2
3 UPDATE accounts
4 SET balance = balance - 200
5 WHERE name = 'Deepak';
6
7 UPDATE accounts
8 SET balance = balance + 200
9 WHERE name = 'Sonu';
10
11 ROLLBACK;
12
13 SELECT * FROM accounts;
```

Text to DDL

Copy as Markdown

Results

There are no results to be displayed.

Query #5 Execution time: 0.21ms

id	name	balance
1	Deepak	5000.0
2	Sonu	3000.0

a)

Schema SQL

```
83 (11,4,390,'completed','2025-01-21 12:00:00'),
84 (12,5,620,'pending','2025-01-22 13:00:00'),
85 (1,6,275,'completed','2025-01-23 14:00:00'),
86 (2,7,480,'completed','2025-01-24 15:00:00'),
87 (3,8,710,'completed','2025-01-25 16:00:00'),
88 (4,1,910,'completed','2025-01-26 17:00:00'),
89 (5,2,260,'completed','2025-01-27 18:00:00'),
90 (6,3,340,'pending','2025-01-28 19:00:00'),
91 (7,4,550,'completed','2025-01-29 20:00:00'),
92 (8,5,430,'completed','2025-01-30 11:30:00'),
93 (9,6,315,'completed','2025-02-01 12:30:00'),
94 (10,7,610,'cancelled','2025-02-02 13:30:00'),
95 (11,8,290,'completed','2025-02-03 14:30:00'),
96 (12,1,410,'completed','2025-02-04 15:30:00');
```

Query SQL

```
1 START TRANSACTION;
2
3 UPDATE accounts
4 SET balance = balance - 200
5 WHERE name = 'Deepak';
6
7 UPDATE accounts
8 SET balance = balance + 200
9 WHERE name = 'Sonu';
10
11 COMMIT;
12
13 SELECT * FROM accounts;
```

Text to DDL

Copy as Markdown

Results

There are no results to be displayed.

Query #5 Execution time: 0.19ms

id	name	balance
1	Deepak	4800.0
2	Sonu	3200.0

b)

c) The ACID property responsible is Atomicity. It ensures that the entire money transfer is treated as a single transaction. If an error occurs before the transfer is completed, all changes are cancelled, preventing any loss or inconsistency in the account balances.

Task 2: Joins

Schema SQL

```
83 (11,4,390,'completed','2025-01-21 12:00:00'),
84 (12,5,620,'pending','2025-01-22 13:00:00'),
85 (1,6,275,'completed','2025-01-23 14:00:00'),
86 (2,7,480,'completed','2025-01-24 15:00:00'),
87 (3,8,710,'completed','2025-01-25 16:00:00'),
88 (4,1,910,'completed','2025-01-26 17:00:00'),
89 (5,2,260,'completed','2025-01-27 18:00:00'),
90 (6,3,340,'pending','2025-01-28 19:00:00'),
91 (7,4,550,'completed','2025-01-29 20:00:00'),
92 (8,5,430,'completed','2025-01-30 11:30:00'),
93 (9,6,315,'completed','2025-02-01 12:30:00'),
94 (10,7,610,'cancelled','2025-02-02 13:30:00'),
95 (11,8,290,'completed','2025-02-03 14:30:00'),
96 (12,1,410,'completed','2025-02-04 15:30:00');
```

Query SQL

```
1 SELECT c.id, c.name, o.id AS order_id, o.total
2 FROM customers c
3 INNER JOIN orders o
4 ON c.id = o.customer_id;
```

Results

id	name	order_id	total
1	Aarav Patel	1	450.0
1	Aarav Patel	11	340.0
1	Aarav Patel	23	275.0
2	Priya Sharma	2	320.0
2	Priya Sharma	12	610.0

a)

Schema SQL

```
83 (11,4,390,'completed','2025-01-21 12:00:00'),
84 (12,5,620,'pending','2025-01-22 13:00:00'),
85 (1,6,275,'completed','2025-01-23 14:00:00'),
86 (2,7,480,'completed','2025-01-24 15:00:00'),
87 (3,8,710,'completed','2025-01-25 16:00:00'),
88 (4,1,910,'completed','2025-01-26 17:00:00'),
89 (5,2,260,'completed','2025-01-27 18:00:00'),
90 (6,3,340,'pending','2025-01-28 19:00:00'),
91 (7,4,550,'completed','2025-01-29 20:00:00'),
92 (8,5,430,'completed','2025-01-30 11:30:00'),
93 (9,6,315,'completed','2025-02-01 12:30:00'),
94 (10,7,610,'cancelled','2025-02-02 13:30:00'),
95 (11,8,290,'completed','2025-02-03 14:30:00'),
96 (12,1,410,'completed','2025-02-04 15:30:00');
```

Query SQL

```
1 SELECT c.id, c.name, o.id AS order_id
2 FROM customers c
3 LEFT JOIN orders o
4 ON c.id = o.customer_id;
```

Results

id	name	order_id
1	Aarav Patel	1
1	Aarav Patel	11
1	Aarav Patel	23
2	Priya Sharma	2

b)

Schema SQL

```
83 (11,4,390,'completed','2025-01-21 12:00:00'),
84 (12,5,620,'pending','2025-01-22 13:00:00'),
85 (1,6,275,'completed','2025-01-23 14:00:00'),
86 (2,7,480,'completed','2025-01-24 15:00:00'),
87 (3,8,710,'completed','2025-01-25 16:00:00'),
88 (4,1,910,'completed','2025-01-26 17:00:00'),
89 (5,2,260,'completed','2025-01-27 18:00:00'),
90 (6,3,340,'pending','2025-01-28 19:00:00'),
91 (7,4,550,'completed','2025-01-29 20:00:00'),
92 (8,5,430,'completed','2025-01-30 11:30:00'),
93 (9,6,315,'completed','2025-02-01 12:30:00'),
94 (10,7,610,'cancelled','2025-02-02 13:30:00'),
95 (11,8,290,'completed','2025-02-03 14:30:00'),
96 (12,1,410,'completed','2025-02-04 15:30:00');
```

Query SQL

```
1 SELECT c.id, c.name
2 FROM customers c
3 LEFT JOIN orders o
4 ON c.id = o.customer_id
5 WHERE o.id IS NULL;
```

Results

id	name
13	Harsh Patel
14	Nikhil Jain
15	Simran Kaur

c) A marketing team would prefer query (b) because it includes customers who have never placed an order, allowing the company to target them with promotions and encourage their first purchase.

Task 3: Indexing

Schema SQL

```
83 (11,4,298, 'completed', '2025-01-21 12:00:00'),
84 (12,5,628, 'pending', '2025-01-22 13:00:00'),
85 (1,6,275, 'completed', '2025-01-23 14:00:00'),
86 (2,7,488, 'completed', '2025-01-24 15:00:00'),
87 (3,8,718, 'completed', '2025-01-25 16:00:00'),
88 (4,1,918, 'completed', '2025-01-26 17:00:00'),
89 (5,2,268, 'completed', '2025-01-27 18:00:00'),
90 (6,3,348, 'pending', '2025-01-28 19:00:00'),
91 (7,4,558, 'completed', '2025-01-29 20:00:00'),
92 (8,5,438, 'completed', '2025-01-30 11:30:00'),
93 (9,6,315, 'completed', '2025-02-01 12:30:00'),
94 (10,7,618, 'cancelled', '2025-02-02 13:30:00'),
95 (11,8,298, 'completed', '2025-02-03 14:30:00'),
96 (12,1,418, 'completed', '2025-02-04 15:30:00');
```

Query SQL

```
1 SHOW INDEX FROM orders;
```

Results

Table	Non_unique	Key_name	Seq_in_index	Column_name	Collation	Cardinality	Sub_part	Packed	Null	Index_type	Comment	Index_comment	Visible	Express
orders	0	PRIMARY	1	id	A	1	null	null		BTREE			YES	null
orders	1	customer_id	1	customer_id	A	1	null	null	YES	BTREE			YES	null
orders	1	restaurant_id	1	restaurant_id	A	1	null	null	YES	BTREE			YES	null

a)

Schema SQL

```
83 (11,4,298, 'completed', '2025-01-21 12:00:00'),
84 (12,5,628, 'pending', '2025-01-22 13:00:00'),
85 (1,6,275, 'completed', '2025-01-23 14:00:00'),
86 (2,7,488, 'completed', '2025-01-24 15:00:00'),
87 (3,8,718, 'completed', '2025-01-25 16:00:00'),
88 (4,1,918, 'completed', '2025-01-26 17:00:00'),
89 (5,2,268, 'completed', '2025-01-27 18:00:00'),
90 (6,3,348, 'pending', '2025-01-28 19:00:00'),
91 (7,4,558, 'completed', '2025-01-29 20:00:00'),
92 (8,5,438, 'completed', '2025-01-30 11:30:00'),
93 (9,6,315, 'completed', '2025-02-01 12:30:00'),
94 (10,7,618, 'cancelled', '2025-02-02 13:30:00'),
95 (11,8,298, 'completed', '2025-02-03 14:30:00'),
96 (12,1,418, 'completed', '2025-02-04 15:30:00');
```

Query SQL

```
1 EXPLAIN ANALYZE
2 SELECT *
3 FROM orders
4 WHERE status = 'completed';
```

Results

Query #1 Execution time: 0.74ms

EXPLAIN

-> Filter ('orders.'status' = 'completed') (cost=3.65 rows=3.4) (actual time=0.0265 0.0388 rows=25 loops=1) -> Table scan on orders (cost=3.65 rows=34) (actual time=0.0207 0.027 rows=34 loops=1)

b)

c) And d)

Schema SQL

```
83 (11,4,298, 'completed', '2025-01-21 12:00:00'),
84 (12,5,628, 'pending', '2025-01-22 13:00:00'),
85 (1,6,275, 'completed', '2025-01-23 14:00:00'),
86 (2,7,488, 'completed', '2025-01-24 15:00:00'),
87 (3,8,718, 'completed', '2025-01-25 16:00:00'),
88 (4,1,918, 'completed', '2025-01-26 17:00:00'),
89 (5,2,268, 'completed', '2025-01-27 18:00:00'),
90 (6,3,348, 'pending', '2025-01-28 19:00:00'),
91 (7,4,558, 'completed', '2025-01-29 20:00:00'),
92 (8,5,438, 'completed', '2025-01-30 11:30:00'),
93 (9,6,315, 'completed', '2025-02-01 12:30:00'),
94 (10,7,618, 'cancelled', '2025-02-02 13:30:00'),
95 (11,8,298, 'completed', '2025-02-03 14:30:00'),
96 (12,1,418, 'completed', '2025-02-04 15:30:00');
```

Query SQL

```
1 CREATE INDEX idx_status
2 ON orders(status);
3 EXPLAIN ANALYZE
4 SELECT *
5 FROM orders
6 WHERE status = 'completed';
```

Results

Query #1 Execution time: 18.88ms

There are no results to be displayed.

Query #2 Execution time: 0.74ms

EXPLAIN

-> Index lookup on orders using idx_status (status='completed') (cost=3.25 rows=25) (actual time=0.049 0.0535 rows=25 loops=1)

e) Initially, the query relied on a full table scan to retrieve the data. Once the index was added, MySQL used the index to access the required records more efficiently, resulting in better query performance.

Task 4: Query Optimization

Schema SQL

```
83 (11,4,290,'completed','2025-01-21 12:00:00'),
84 (12,5,620,'pending','2025-01-22 13:00:00'),
85 (1,6,275,'completed','2025-01-23 14:00:00'),
86 (2,7,400,'completed','2025-01-24 15:00:00'),
87 (3,8,710,'completed','2025-01-25 16:00:00'),
88 (4,1,910,'completed','2025-01-26 17:00:00'),
89 (5,2,260,'completed','2025-01-27 18:00:00'),
90 (6,3,340,'pending','2025-01-28 19:00:00'),
91 (7,4,550,'completed','2025-01-29 20:00:00'),
92 (8,5,430,'completed','2025-01-30 11:30:00'),
93 (9,6,315,'completed','2025-02-01 12:30:00'),
94 (10,7,610,'cancelled','2025-02-02 13:30:00'),
95 (11,8,290,'completed','2025-02-03 14:30:00'),
96 (12,1,410,'completed','2025-02-04 15:30:00');
```

Text to DDL

Query SQL

```
1 SELECT *
2 FROM orders
3 WHERE customer_id IN (
4     SELECT id
5     FROM customers
6     WHERE signup_date >= '2024-04-01'
7 );
```

Results

Copy as Markdown

Query #1 Execution time: 36.88ms

id	customer_id	restaurant_id	total	status	created_at
8	8	8	290.0	pending	2025-01-08 19:00:00
18	8	1	330.0	completed	2025-01-18 18:00:00
30	8	5	430.0	completed	2025-01-30 11:30:00
9	9	1	180.0	completed	2025-01-09 12:30:00

a)

Schema SQL

```
13 (11,4,290,'completed','2025-01-21 12:00:00'),
14 (12,5,620,'pending','2025-01-22 13:00:00'),
15 (1,6,275,'completed','2025-01-23 14:00:00'),
16 (2,7,400,'completed','2025-01-24 15:00:00'),
17 (3,8,710,'completed','2025-01-25 16:00:00'),
18 (4,1,910,'completed','2025-01-26 17:00:00'),
19 (5,2,260,'completed','2025-01-27 18:00:00'),
20 (6,3,340,'pending','2025-01-28 19:00:00'),
21 (7,4,550,'completed','2025-01-29 20:00:00'),
22 (8,5,430,'completed','2025-01-30 11:30:00'),
23 (9,6,315,'completed','2025-02-01 12:30:00'),
24 (10,7,610,'cancelled','2025-02-02 13:30:00'),
25 (11,8,290,'completed','2025-02-03 14:30:00'),
26 (12,1,410,'completed','2025-02-04 15:30:00');
```

Text to DDL

Query SQL

```
1 SELECT o.*
2 FROM orders o
3 JOIN customers c
4 ON o.customer_id = c.id
5 WHERE c.signup_date >= '2024-04-01';
```

Results

Copy as Markdown

Query #1 Execution time: 0.85ms

id	customer_id	restaurant_id	total	status	created_at
8	8	8	290.0	pending	2025-01-08 19:00:00
18	8	1	330.0	completed	2025-01-18 18:00:00
30	8	5	430.0	completed	2025-01-30 11:30:00
9	9	1	180.0	completed	2025-01-09 12:30:00

b)

Schema SQL

```
83 (11,4,290,'completed','2025-01-21 12:00:00'),
84 (12,5,620,'pending','2025-01-22 13:00:00'),
85 (1,6,275,'completed','2025-01-23 14:00:00'),
86 (2,7,400,'completed','2025-01-24 15:00:00'),
87 (3,8,710,'completed','2025-01-25 16:00:00'),
88 (4,1,910,'completed','2025-01-26 17:00:00'),
89 (5,2,260,'completed','2025-01-27 18:00:00'),
90 (6,3,340,'pending','2025-01-28 19:00:00'),
91 (7,4,550,'completed','2025-01-29 20:00:00'),
92 (8,5,430,'completed','2025-01-30 11:30:00'),
93 (9,6,315,'completed','2025-02-01 12:30:00'),
94 (10,7,610,'cancelled','2025-02-02 13:30:00'),
95 (11,8,290,'completed','2025-02-03 14:30:00'),
96 (12,1,410,'completed','2025-02-04 15:30:00');
```

Text to DDL

Query SQL

```
1 SELECT
2 o.id,
3 c.name,
4 o.total,
5 o.status,
6 o.created_at
7 FROM orders o
8 JOIN customers c
9 ON o.customer_id = c.id
10 WHERE c.signup_date >= '2024-04-01';
```

Results

Copy as Markdown

Query #1 Execution time: 0.9ms

id	name	total	status	created_at
8	Anjali Shah	290.0	pending	2025-01-08 19:00:00
18	Anjali Shah	330.0	completed	2025-01-18 18:00:00
30	Anjali Shah	430.0	completed	2025-01-30 11:30:00
9	Karan Desai	180.0	completed	2025-01-09 12:30:00

c)

Task 5:

Schema SQL

```
83 (11,4,390,'completed','2025-01-21 12:00:00'),
84 (12,5,620,'pending','2025-01-22 13:00:00'),
85 (1,6,275,'completed','2025-01-23 14:00:00'),
86 (2,7,480,'completed','2025-01-24 15:00:00'),
87 (3,8,710,'completed','2025-01-25 16:00:00'),
88 (4,1,910,'completed','2025-01-26 17:00:00'),
89 (5,2,260,'completed','2025-01-27 18:00:00'),
90 (6,3,340,'pending','2025-01-28 19:00:00'),
91 (7,4,550,'completed','2025-01-29 20:00:00'),
92 (8,5,430,'completed','2025-01-30 11:30:00'),
93 (9,6,315,'completed','2025-02-01 12:30:00'),
94 (10,7,610,'cancelled','2025-02-02 13:30:00'),
95 (11,8,290,'completed','2025-02-03 14:30:00'),
96 (12,1,410,'completed','2025-02-04 15:30:00');
```

Text to DDL

Query SQL

```
1 SELECT
2   r.name AS restaurant_name,
3   o.total,
4   o.status,
5   o.created_at
6 FROM orders o
7 JOIN restaurants r
8   ON o.restaurant_id = r.id
9 WHERE o.customer_id = 1
10 ORDER BY o.created_at DESC
11 LIMIT 5;
```

Results

Copy as Markdown

Query #1 Execution time: 0.81ms

restaurant_name	total	status	created_at
Sushi House	275.0	completed	2025-01-23 14:00:00
Burger Point	340.0	completed	2025-01-11 11:00:00
Pizza Hub	450.0	completed	2025-01-01 12:00:00

a)

Schema SQL

```
83 (11,4,390,'completed','2025-01-21 12:00:00'),
84 (12,5,620,'pending','2025-01-22 13:00:00'),
85 (1,6,275,'completed','2025-01-23 14:00:00'),
86 (2,7,480,'completed','2025-01-24 15:00:00'),
87 (3,8,710,'completed','2025-01-25 16:00:00'),
88 (4,1,910,'completed','2025-01-26 17:00:00'),
89 (5,2,260,'completed','2025-01-27 18:00:00'),
90 (6,3,340,'pending','2025-01-28 19:00:00'),
91 (7,4,550,'completed','2025-01-29 20:00:00'),
92 (8,5,430,'completed','2025-01-30 11:30:00'),
93 (9,6,315,'completed','2025-02-01 12:30:00'),
94 (10,7,610,'cancelled','2025-02-02 13:30:00'),
95 (11,8,290,'completed','2025-02-03 14:30:00'),
96 (12,1,410,'completed','2025-02-04 15:30:00');
```

Text to DDL

Query SQL

```
1 EXPLAIN ANALYZE
2 SELECT
3   r.name AS restaurant_name,
4   o.total,
5   o.status,
6   o.created_at
7 FROM orders o
8 JOIN restaurants r
9   ON o.restaurant_id = r.id
10 WHERE o.customer_id = 1
11 ORDER BY o.created_at DESC
12 LIMIT 5;
```

Results

Copy as Markdown

Query #1 Execution time: 1.87ms

EXPLAIN

-> Limit: 5 row(s) (cost=2.1 rows=3) (actual time=0.062..0.0661 rows=3 loops=1) -> Nested loop inner join (cost=2.1 rows=3) (actual time=0.0586..0.0625 rows=3 loops=1) -> Sort: o.created_at DESC (cost=1.05 rows=3) (actual time=0.0422..0.0426 rows=3 loops=1) -> Filter: (o.restaurant_id is not null) (cost=1.05 rows=3) (actual time=0.0302..0.0323 rows=3 loops=1) -> Index lookup on o using customer_id (customer_id=1) (cost=1.05 rows=3) (actual time=0.0293..0.0311 rows=3 loops=1) -> Single-row index lookup on r using PRIMARY (id=o.restaurant_id) (cost=0.283 rows=1) (actual time=0.00603..0.00608 rows=1 loops=3)

b)

c) And d)

Schema SQL

```
83 (11,4,390,'completed','2025-01-21 12:00:00'),
84 (12,5,620,'pending','2025-01-22 13:00:00'),
85 (1,6,275,'completed','2025-01-23 14:00:00'),
86 (2,7,480,'completed','2025-01-24 15:00:00'),
87 (3,8,710,'completed','2025-01-25 16:00:00'),
88 (4,1,910,'completed','2025-01-26 17:00:00'),
89 (5,2,260,'completed','2025-01-27 18:00:00'),
90 (6,3,340,'pending','2025-01-28 19:00:00'),
91 (7,4,550,'completed','2025-01-29 20:00:00'),
92 (8,5,430,'completed','2025-01-30 11:30:00'),
93 (9,6,315,'completed','2025-02-01 12:30:00'),
94 (10,7,610,'cancelled','2025-02-02 13:30:00'),
95 (11,8,290,'completed','2025-02-03 14:30:00'),
96 (12,1,410,'completed','2025-02-04 15:30:00');
```

Text to DDL

Query SQL

```
1 CREATE INDEX idx_customer_created
2 ON orders(customer_id, created_at);
3 EXPLAIN ANALYZE
4 SELECT
5   r.name AS restaurant_name,
6   o.total,
7   o.status,
8   o.created_at
9 FROM orders o
10 JOIN restaurants r
11   ON o.restaurant_id = r.id
12 WHERE o.customer_id = 1
13 ORDER BY o.created_at DESC
14 LIMIT 5;
```

Results

Copy as Markdown

Query #1 Execution time: 9.2ms

There are no results to be displayed.

Query #2 Execution time: 8.73ms

EXPLAIN

-> Limit: 5 row(s) (cost=2.1 rows=3) (actual time=0.0304..0.0373 rows=3 loops=1) -> Nested loop inner join (cost=2.1 rows=3) (actual time=0.0266..0.0332 rows=3 loops=1) -> Filter: (o.restaurant_id is not null) (cost=1.05 rows=3) (actual time=0.0187..0.0217 rows=3 loops=1) -> Index lookup on o using idx_customer_created (customer_id=1) (reverse) (cost=1.05 rows=3) (actual time=0.018..0.0206 rows=3 loops=1) -> Single-row index lookup on r using PRIMARY (id=o.restaurant_id) (cost=0.283 rows=1) (actual time=0.00309..0.00313 rows=1 loops=3)

e)

Schema SQL

```
83 (11,4,398,'completed','2025-01-21 12:00:00'),
84 (12,5,620,'pending','2025-01-22 13:00:00'),
85 (1,6,275,'completed','2025-01-24 14:00:00'),
86 (2,7,480,'completed','2025-01-24 15:00:00'),
87 (3,8,710,'completed','2025-01-25 16:00:00'),
88 (4,1,910,'completed','2025-01-26 17:00:00'),
89 (5,2,260,'completed','2025-01-27 18:00:00'),
90 (6,3,340,'pending','2025-01-28 19:00:00'),
91 (7,4,550,'completed','2025-01-29 20:00:00'),
92 (8,5,430,'completed','2025-01-30 11:30:00'),
93 (9,6,315,'completed','2025-02-01 12:30:00'),
94 (10,7,610,'cancelled','2025-02-02 13:30:00'),
95 (11,8,290,'completed','2025-02-03 14:30:00'),
96 (12,1,410,'completed','2025-02-04 15:30:00');
```

Text to DDL

Query SQL

```
1 SELECT
2   r.name,
3   o.total,
4   o.status,
5   o.created_at
6 FROM orders o
7 JOIN restaurants r
8 ON o.restaurant_id = r.id
9 WHERE o.customer_id = 1
10 ORDER BY o.created_at DESC
11 LIMIT 5;
```

Results

Copy as Markdown

Query #1 Execution time: 1.15ms

name	total	status	created_at
Sushi House	275.0	completed	2025-01-23 14:00:00
Burger Point	340.0	completed	2025-01-11 11:00:00
Pizza Hub	450.0	completed	2025-01-01 12:00:00

f) Initially, MySQL took longer because it had to examine more data while processing the query. Once the index was created, the database accessed the required rows more quickly, making the query faster and more efficient.

